V SEMESTER (OPEN ELECTIVE –I)	L	T	P	C
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20CS5O02 - DATA ENGINEERING				

COURSE OUTCOMES:
Student able to state and analyze:
CO1: Preprocessing techniques for various datasets,
CO2: Standard database systems concepts like tables, relations, query,NoSQL
CO3: Information retrieval techniques such as Relevance Ranking,Indexing etc
CO4: Data processing algorithms and data structures
CO5: Visualization techniques like Table, graph, histogram, pie-chart

UNIT- I:
Data Engineering:introduction, importance of data engineering, Data engineering vs data science
Data Collection: Various sources of data, types of data: text, video, audio, biology etc.
Data Preprocessing: data Cleaning: missing values, noise elimination, data integration, data transformation: Normalization, Data Reduction: data cube aggregation, dimensionality reduction.

UNIT-II
Data bases: Database Schema, ER diagram, introduction to SQL, functions and stored procedures, indexing: B+tree index files, data base system architecture: Client-Server Architecture, introduction to MongoDB,
NoSQL: The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, Standard Model, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL

UNIT-III
Information Retrieval: Relevance Ranking Using Terms,Relevance Using Hyperlinks,Synonyms, Homonyms, and Ontologies,Indexing of Documents,Measuring Retrieval Effectiveness,Crawling and Indexing the Web,Information Retrieval: Beyond Ranking of Pages, Directories and Categories

UNIT-IV
Data Analysis: correlation analysis: bivariate correlation, partial correlation, correlation coefficients. Regression: simple linear regression, multiple linear regression, principal component analysis, analysis of variance

UNIT- V
Data Visualization: Table, graph, histogram, pie-chart, area-plot, box-plot, scatter-plot, bubble-plot, waffle charts, word clouds.

Text Books:
1. Data mining concepts and techniques Jiawei Han and Micheline Kamber (UNIT-I)
2. Silberschatz, Korth, Sudarshan, “Database System Concepts” McGraw Hill Education, Sixth edition, 2010, ISBN-13: 978-9332901384 (UNIT-II, UNIT-III)
3. “NoSQL distilled” A Brief Guide to the Emerging World of Polyglot PersistencePramod J. Sadalage Martin Fowler, Addison Wesley (UNIT-II)
4. correlation and regression analysis by Dr. Mohamed Ahmed Zaid (UNIT-IV)

Reference Books:
1. Brian Shive, “Data Engineering: A Novel Approach to Data Design”, Technics Publications, 2013. ISBN-13: 978-1935504603.
2. Joel Grus, “Python Data Science Handbook: Essential Tools for Working with Data”, 1st Edition, O’Reilly, 2016. ISBN-13: 978-9352134915.

Web links:
1. https://chartio.com/learn/charts/essential-chart-types